

$$1. (1) \operatorname{div} \vec{A} = 8xy + 3y^2$$

5'

$$\operatorname{curl} \vec{A} = 4xz \vec{x}_0 + (1-2yz) \vec{y}_0 + (-3x^2 - z^2) \vec{z}_0$$

5'

$$(2) \operatorname{div} \vec{A} = 0$$

5'

$$\operatorname{curl} \vec{A} = (2xy - x^2) \vec{x}_0 + (2yz - y^2) \vec{y}_0 + (2xz - z^2) \vec{z}_0$$

5'

$$(3) \operatorname{div} \vec{A} = 1 + 2y$$

5'

$$\operatorname{curl} \vec{A} = (2x-1) \vec{z}_0$$

5'

2. 不能.

15'

$$\text{切面 } \vec{E} = 0 \quad \nRightarrow \quad \nabla \times \vec{E} = 0$$

$$3. \vec{A} \cdot \vec{B} = 5 - 3j$$

10'

$$\vec{A} \times \vec{B} = (-1 + 6j) \vec{x}_0 + (1 + 3j) \vec{y}_0 + (-2 - 3j) \vec{z}_0$$

10'

$$\vec{A} \cdot \vec{B}^* = -3 - 0j$$

10'

$$\operatorname{Re}(\vec{A} \times \vec{B}^*) = 5 \vec{x}_0 + \vec{y}_0 - 2 \vec{z}_0$$

10'

$$4. \vec{n} = (2, 2, 1) / (-2, -2, 1)$$

15'

2. ①

$$\vec{H} = \frac{1}{\mu} \vec{B}, \quad \vec{B} = \epsilon \vec{E}$$

$$\nabla \times \vec{H} = \vec{J} + j\omega \vec{D}$$

$$\frac{1}{\mu} \nabla \times \vec{B} = \vec{J} + j\omega \epsilon \vec{E}$$

$$\frac{1}{j\omega \epsilon} \left(\frac{1}{\mu} \nabla \times \vec{B} - \vec{J} \right) = \vec{E}$$

如果 $\vec{E} = 0$,

$$\text{则 } \frac{1}{\mu} \nabla \times \vec{B} = \vec{J}$$

可能还存在传导电流。

2. ②

$$\text{假定 } \vec{E} = y \vec{x}$$

$$\frac{\partial \vec{B}}{\partial t} = -\nabla \times \vec{E} = \vec{z}$$

在 $y=0$ 平面 $E=0$, 但 $\frac{\partial B}{\partial t} \neq 0$.